Automatic-Censoring K-Distribution CFAR Detection for Active Sonar in Heavy-Tailed Reverberation

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***Abstract*—Constant false alarm rate (CFAR) detectors used in active sonar are usually designed for Gaussian backgrounds, yet sonar reverberation is heavy-tailed, and reference windows are further corrupted by interfering targets and reverberation edges. We study CFAR behaviour under K-distributed reverberation and propose ACK-CFAR, which combines a log-moment estimate of the K shape parameter with model-based automatic censoring and an estimator-aware threshold. Monte Carlo simulations show that CA-, OS- and VI-CFAR exceed the design false-alarm probability by roughly 16–25 times in spiky homogeneous reverberation, while the calibrated ACK-CFAR reduces the excess to about 3 times, and to about 1 time in the presence of three interfering targets. The price is a detection loss of 1–2 dB relative to classical detectors at matched false-alarm probability, caused mainly by shape estimation from a short window. We discuss this trade-off and outline how longer-window shape estimation may close the gap.**

***Index Terms*—active sonar, CFAR, K-distribution, reverberation, automatic censoring**

I. Introduction

Active sonar detection relies on adaptive thresholds that keep the false-alarm rate constant as the background changes. Classical mean-level and order-statistic detectors (CA, GO, SO, OS) [1], [2] assume a Gaussian (exponential-intensity) background. In practice, sonar reverberation departs from the Rayleigh model toward heavy-tailed distributions such as log-normal, Weibull and K [5]. K-distribution based CFAR has been proposed for sonar imagery [4], variability-index CFAR has been extended to multistatic sonar [6], and recent work combines segmentation with local CFAR [7].

We ask a narrower question: how do standard detectors behave when heavy-tailed reverberation, interfering targets and reverberation edges occur together, and can a K-model detector with automatic censoring control false alarms better? Our contributions are (i) a controlled simulation comparison of CA, GO, SO, OS, VI and K-model detectors in these conditions, (ii) ACK-CFAR, and (iii) an honest account of the detection-loss cost.

II. Signal Model

The reverberation intensity in a cell is $I = \tau E$, where $E \sim \mathrm{Exp}(1)$ is speckle and $\tau \sim \mathrm{Gamma}(\nu, \mu/\nu)$ is texture, giving the K-distribution with shape ν and mean μ. The tail is

$$P(I > t) = \frac{2}{\Gamma(\nu)} \left(\frac{\nu t}{\mu} \right)^{\nu/2} K_\nu \left(2\sqrt{\nu t/\mu} \right). \qquad (1)$$

Small ν means spiky reverberation; $\nu \to \infty$ recovers the Gaussian case. A Swerling-I target of signal-to-clutter ratio (SCR) adds $\mathrm{SCR} \cdot \mu$ to the texture of the cell under test (CUT). We use a reference window of 2×16 cells with 2 guard cells per side.

III. Detectors

A. Baselines

CA, GO, SO and OS ($k = 3N/4$) thresholds are designed for exponential background (scale factors by closed form or Monte Carlo). VI-CFAR switches among CA, GO and SO using variability-index and mean-ratio tests [3]. K-LM estimates (ν, μ) by log-moments without censoring.

B. Proposed ACK-CFAR

Since $\mathrm{Var}[\ln I] = \psi_1(\nu) + \pi^2/6$, the shape is obtained by inverting the trigamma function on the sample variance of $\ln I$ over retained reference cells. Cells above $c(\hat{\nu})\hat{\mu}$ are censored, where $c(\cdot)$ is the K-tail quantile at a small probability p_c; because c grows for spiky reverberation, genuine clutter spikes are kept while strong interferers are removed. The mean of retained cells is corrected for truncation, and the procedure is iterated twice. ACK-CFAR uses the analytic K threshold $\kappa(\hat{\nu})\hat{\mu}$; ACK-cal replaces κ by a Monte Carlo learned function of $\hat{\nu}$, accounting for estimation noise.

IV. Simulation Setup

Scenarios: homogeneous reverberation; three interferers at 15 dB inside the reference window; and a 10 dB reverberation edge with the CUT on the high or low side. Shape values $\nu \in \{1, 3, \infty\}$. Design $P_{fa} = 10^{-3}$, 10^6 trials for P_{fa} and 2×10^4 per SCR point for P_d. Detectors are also compared at *matched* P_{fa} by setting each detector's threshold to its own empirical $(1 - P_{fa})$ quantile. An oracle with known (ν, μ) gives a bound. Code is available from the authors.

V. Results

Fig. 1 shows P_d against SCR at matched P_{fa} for $\nu = 1$, Fig. 2 the measured P_{fa} for all detectors, and Table I the measured P_{fa} (as a multiple of the design value) together with the SCR required for $P_d = 0.9$ at matched P_{fa}.
Oracle reference: $P_{fa} \approx 1$ and SCR $\approx$ 22.3 dB in the homogeneous case. In Gaussian noise ($\nu = \infty$) all classical detectors hold the design P_{fa}, while ACK-cal is conservative.

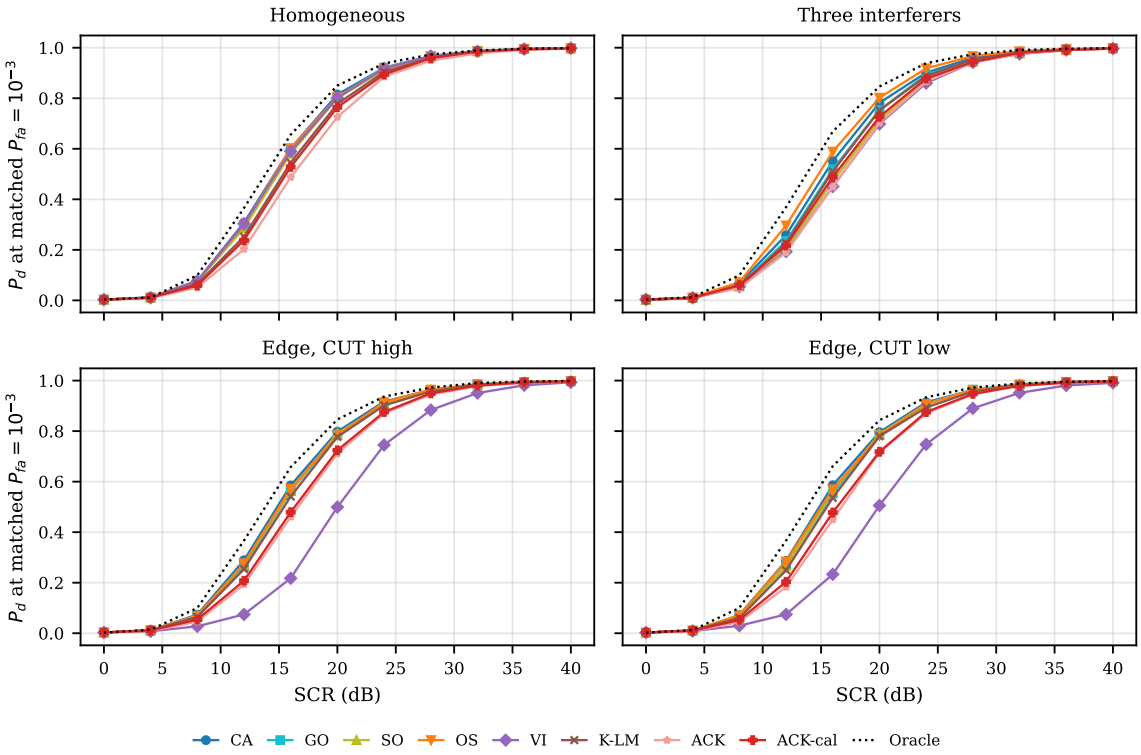


Fig. 1. P_d versus SCR at matched $P_{fa} = 10^{-3}$ for $\nu = 1$ in the four scenarios.

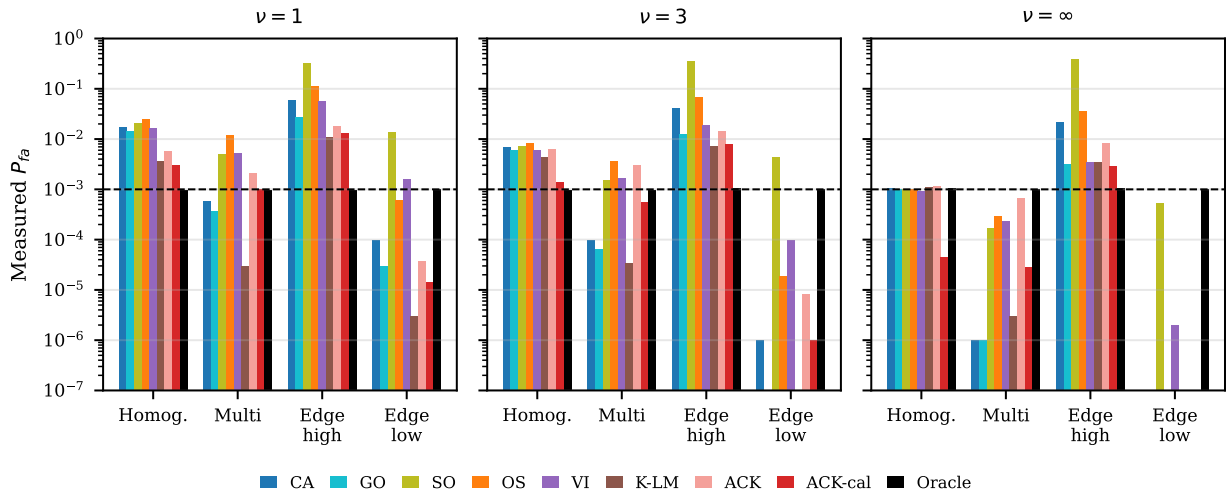


Fig. 2. Measured P_{fa} against the design value 10^{-3} (dashed) for all detectors and scenarios.

Classical detectors designed for Gaussian noise exceed the design false-alarm rate by more than an order of magnitude in spiky reverberation. ACK-cal substantially reduces the excess, but at matched P_{fa} it loses 1–2 dB against CA-type detectors, and more when the CUT lies on the low side of an edge. Most of the remaining gap to the oracle is due to estimating ν from 32 cells.

VI. Discussion and Limitations

The advantage demonstrated is false-alarm control, not detection gain. The evaluation is synthetic with independent texture across cells, whereas real reverberation has correlated texture; K-LM is a simplified stand-in for published K-CFAR detectors; and no real sonar data were used. Future work: estimate shape over a long window or in 2-D, add explicit edge logic, and validate on measured data.

TABLE I
MEASURED P_{fa} AS A MULTIPLE OF THE DESIGN VALUE (10^{-3}) / SCR (DB) FOR $P_d = 0.9$ AT MATCHED P_{fa}

ν	Scenario	CA	GO	SO	OS	VI	K-LM	ACK	ACK-cal	Oracle
1	Homog.	17/23.2	14/23.3	21/23.6	24/23.4	16/23.4	3.7/23.9	5.8/24.9	3.1/24.4	1.0/22.3
	3 interf.	0.6/24.0	0.4/24.5	4.9/25.6	12/23.4	5.1/25.9	0.0/24.8	2.0/25.8	1.0/25.3	1.0/22.3
	Edge hi	59/23.4	27/23.8	314/23.8	112/23.6	55/29.0	11/23.9	18/25.6	13/25.3	0.9/22.4
	Edge lo	0.1/23.5	0.0/23.8	13/23.7	0.6/23.8	1.5/28.6	0.0/24.4	0.0/25.6	0.0/25.4	1.0/22.5
3	Homog.	6.8/21.0	6.0/21.2	7.3/21.3	8.0/21.3	6.1/21.1	4.4/22.1	6.4/22.7	1.4/21.9	1.0/20.1
	3 interf.	0.1/22.6	0.1/23.1	1.5/23.3	3.6/21.4	1.7/23.6	0.0/23.8	2.9/23.3	0.6/22.5	1.0/20.3
	Edge hi	40/21.5	12/21.6	353/21.7	68/22.1	18/24.5	7.2/22.6	14/23.9	8.0/23.2	1.1/20.4
	Edge lo	0.0/21.4	0.0/21.6	4.4/21.7	0.0/22.0	0.1/24.2	0.0/22.6	0.0/23.8	0.0/23.3	1.0/20.1
∞	Homog.	1.1/18.9	1.0/19.1	1.0/19.2	1.0/19.1	0.9/18.9	1.1/19.4	1.2/19.7	0.0/19.2	1.0/18.5
	3 interf.	0.0/21.0	0.0/21.7	0.2/21.8	0.3/19.1	0.2/22.1	0.0/23.2	0.6/20.3	0.0/19.7	1.0/18.4
	Edge hi	22/19.2	3.2/19.3	394/19.3	35/19.8	3.4/19.5	3.4/20.4	8.3/22.5	2.8/21.5	1.1/18.5
	Edge lo	0.0/19.2	0.0/19.3	0.5/19.4	0.0/19.9	0.0/19.5	0.0/20.5	0.0/22.5	0.0/21.4	1.0/18.4

VII. CONCLUSION

Heavy-tailed reverberation makes Gaussian-designed CFAR detectors unreliable in false-alarm control. Automatic censoring with a log-moment K model restores much of it at a modest detection cost, and shape-estimation accuracy is the main factor to improve.

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